PLSC 30600

Week 8: Regression discontinuity (catch-up) + differences in differences

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Roadmap

- Part A (catch-up): regression discontinuity (RD)
- Part B: differences in differences (DiD) and panel-data designs (next)

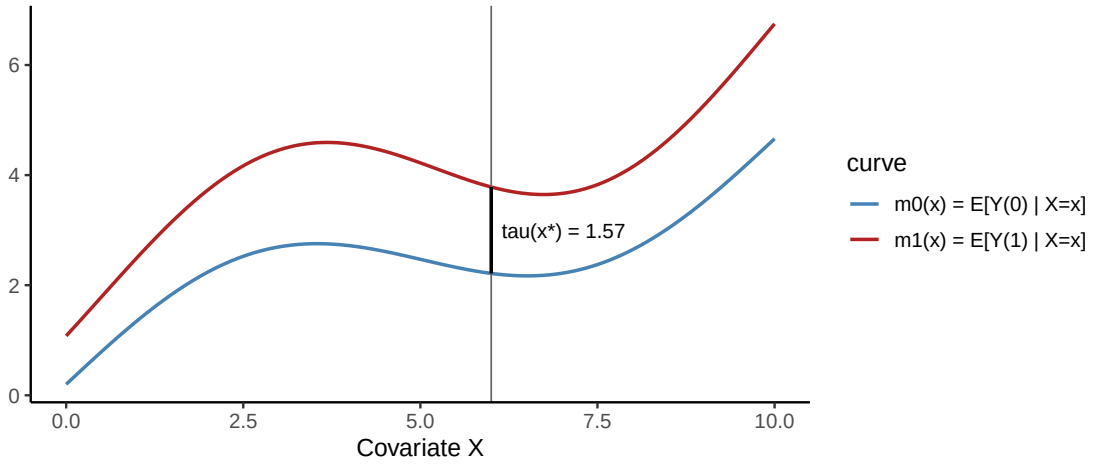
Potential outcomes as functions of X

Think of the average potential outcomes as smooth regression functions:

$$m_0(x) \equiv \mathrm{E}\left[Y_i(0) \mid X_i = x\right], \qquad m_1(x) \equiv \mathrm{E}\left[Y_i(1) \mid X_i = x\right].$$

The (conditional) average treatment effect at a point $x^\star$ is

$$\tau(x^\star) \equiv \mathrm{E}\left[Y_i(1) - Y_i(0) \mid X_i = x^\star\right] = m_1(x^\star) - m_0(x^\star).$$



RD setup (Imbens–Lemieux notation)

We observe (Y_i, D_i, X_i), where X_i is the forcing variable and $D_i \in \{0, 1\}$ is treatment/receipt.

- Cutoff: c.
- Core comparison: units just below vs. just above c (*local*).
- **Sharp RD (SRD):** $D_i = \mathbf{1}\{X_i \geq c\}$.
- **Fuzzy RD (FRD):** $\Pr[D_i = 1 \mid X_i = x]$ jumps at c, but not from 0 to 1.

FRD is a local IV design: crossing the cutoff shifts treatment probability.

Many notes re-center the forcing variable so the cutoff is 0: define $V_i \equiv X_i - c$. We mainly write X_i and c.

Motivation pictures: assignment and regression functions

In SRD, the treatment probability jumps from 0 to 1 at the cutoff, so the observed outcome regression function inherits a jump.

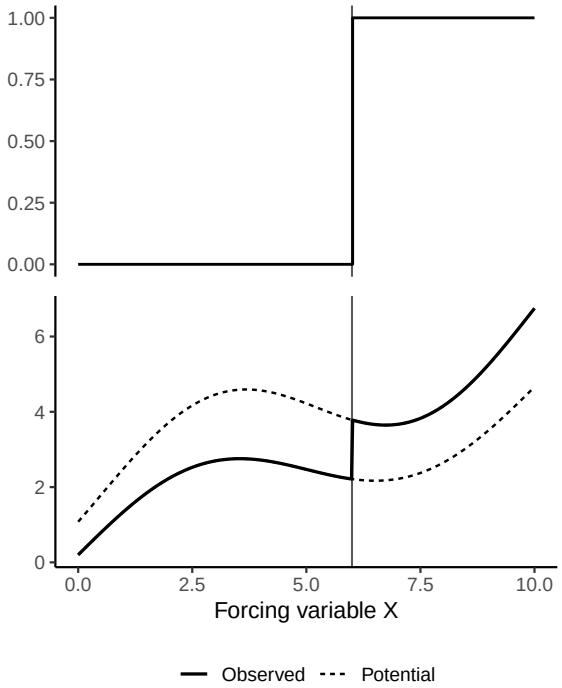


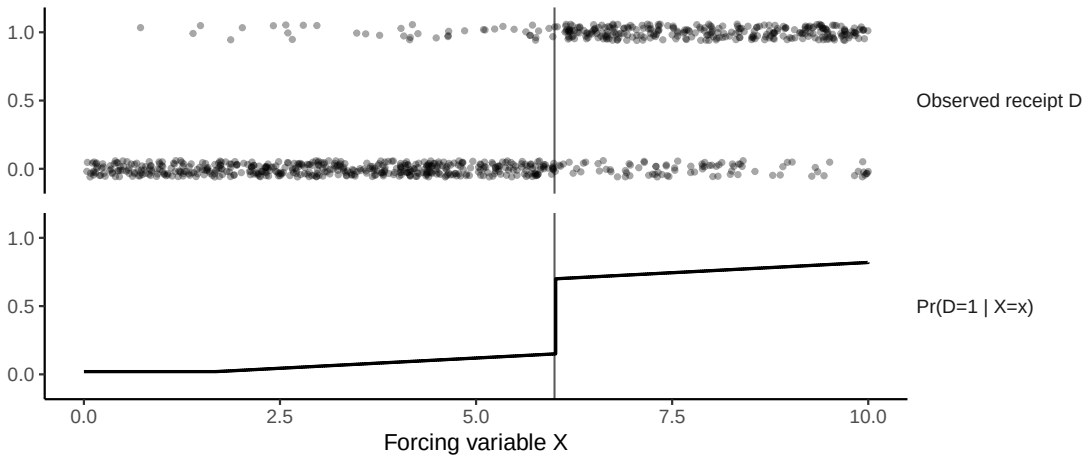
Fig. 1. Assignment probabilities (SRD).

Fig. 2. Potential and observed outcome regression functions.

Motivating FRD: imprecise sorting and imperfect compliance

Often there is a cutoff-based **eligibility rule** but **treatment receipt is imperfect**.

- Define eligibility (the instrument): $Z_i = \mathbf{1}\{X_i \geq c\}$.
- With imprecise sorting / imperfect enforcement / imperfect take-up, D_i does not equal Z_i for everyone.
- Then $\Pr[D_i = 1 \mid X_i = x]$ still has a *jump* at c, but not from 0 to 1 (fuzzy RD).



RD assumptions (sufficient conditions, in our notation)

We observe (Y_i, D_i, X_i), where X_i is the forcing variable and $D_i \in \{0, 1\}$ is treatment/receipt.

- **SUTVA:** $Y_i = Y_i(1) \cdot D_i + Y_i(0) \cdot (1 - D_i)$.

- **Continuity of potential-outcome CEFs at the cutoff:** for $d \in \{0, 1\}$,

$$\lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i(d) \mid X_i = c + \ell\right] = \lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i(d) \mid X_i = c - \ell\right] = \mathrm{E}\left[Y_i(d) \mid X_i = c\right].$$

- **Density/positivity near the cutoff:** f_X is continuous at c and $f_X(c) > 0$ (so there are observations arbitrarily close to c on both sides).

Strictly speaking, continuity of f_X is not required for identification, but it is often a useful (and checkable) diagnostic: big density changes can signal sorting/manipulation.

SRD identification: the jump equals the ATE at the cutoff

Define the SRD estimand (Imbens–Lemieux):

$$\tau_{SRD} \equiv \mathrm{E}\left[Y_i(1) - Y_i(0) \mid X_i = c\right].$$

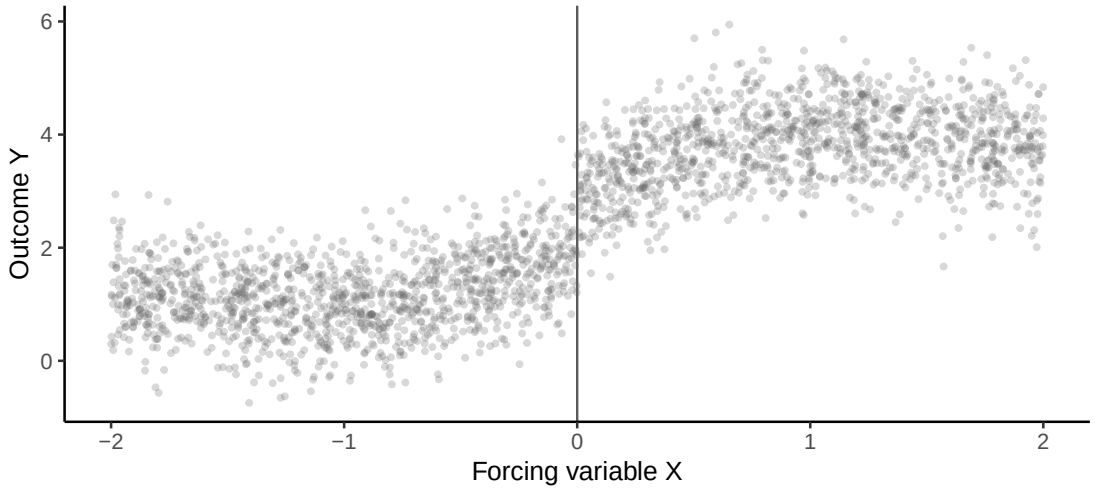
In **sharp RD**, $D_i = \mathbf{1}\{X_i \geq c\}$. Using SUTVA and continuity at c:

$$\begin{aligned}
&\lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i \mid X_i = c + \ell\right] - \lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i \mid X_i = c - \ell\right] \\
&= \lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i(1) \mid X_i = c + \ell\right] - \lim_{\ell \downarrow 0} \mathrm{E}\left[Y_i(0) \mid X_i = c - \ell\right] \\
&= \mathrm{E}\left[Y_i(1) \mid X_i = c\right] - \mathrm{E}\left[Y_i(0) \mid X_i = c\right] \\
&= \mathrm{E}\left[Y_i(1) - Y_i(0) \mid X_i = c\right] \\
&\equiv \tau_{SRD}.
\end{aligned}$$

Interpretation: the jump in $\mathrm{E}\left[Y \mid X = x\right]$ at $x = c$ equals the ATE for units at the cutoff.

Simulated toy RD: raw data

Before smoothing or binning, the data are noisy: you cannot read the jump off of individual points.



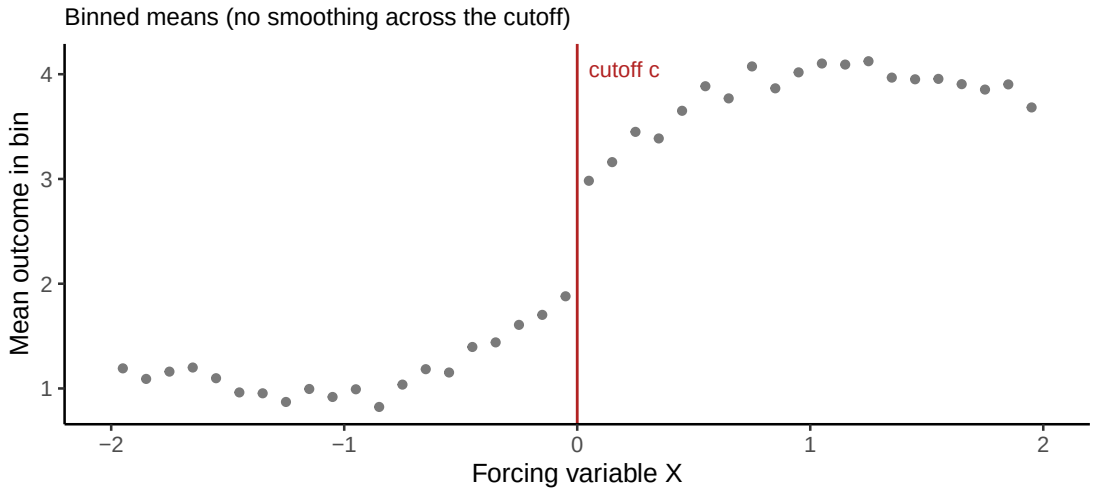
Step 1 in practice: a binned scatter plot

Plot bin means of Y against X, separately on each side of the cutoff (do not smooth across c).

If the plot shows many jumps of similar size away from c, interpreting the jump at c as causal is harder.

Simulated toy RD: binned means

By construction, the true cutoff effect is $\tau_{SRD} = 1$ in this toy example.



Estimation: local linear regression at a boundary

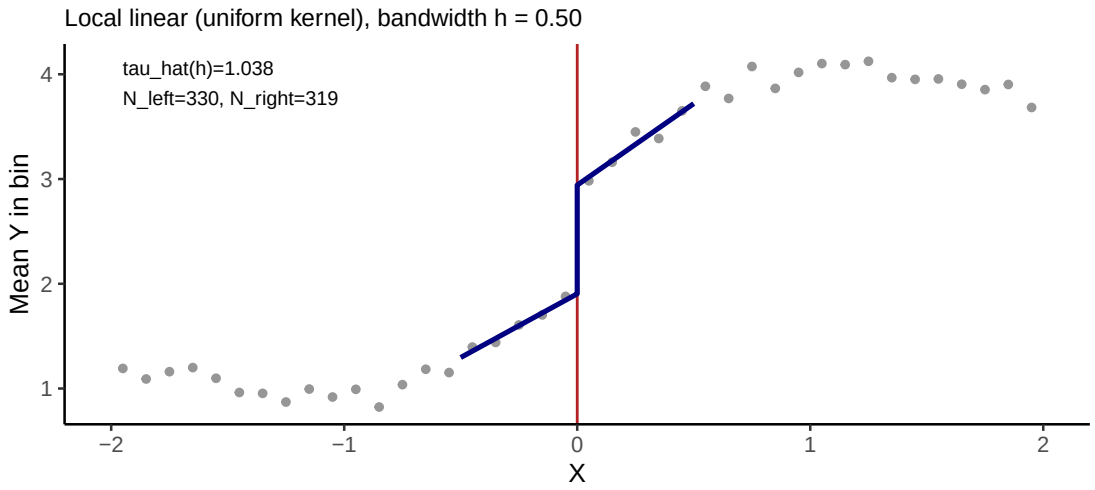
Imbens–Lemieux recommendation: local linear regression, fit separately on each side using only $|X_i - c| \leq h$.

- With a uniform kernel, this is "OLS within bandwidth h" on each side.
- Estimate $\hat{\tau}(h)$ as the difference between the fitted intercepts at c.

Key practice point: report robustness to multiple bandwidths.

Toy RD: local linear fit for one bandwidth

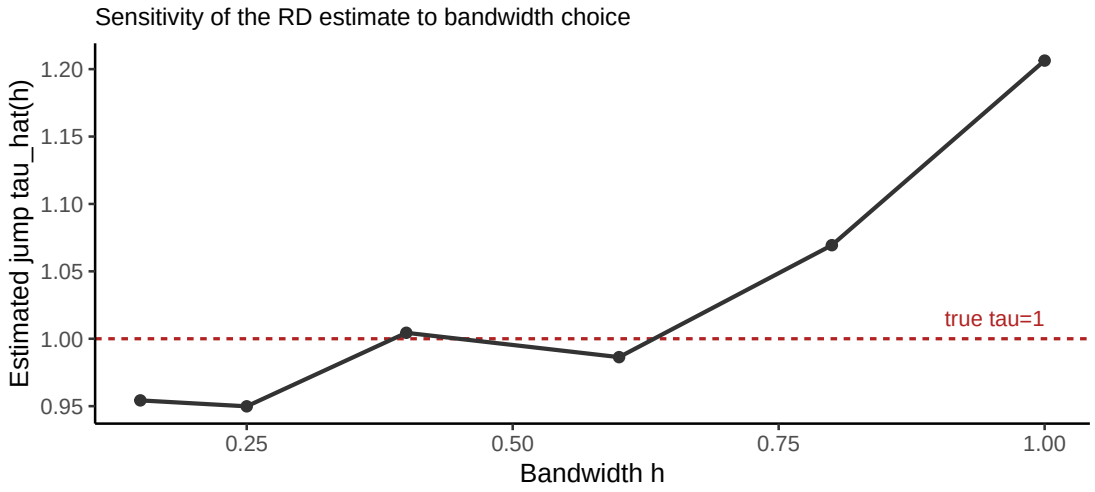
Example with bandwidth $h = 0.5$: fit a line to Y vs. $(X - c)$ on each side using only $|X - c| \leq h$.



Toy RD: bandwidth sensitivity

Compute $\hat{\tau}(h)$ for a range of bandwidths h.

Data-driven "optimal" bandwidth selectors are useful, but you should still report sensitivity; see Imbens and Kalyanaraman (2012).



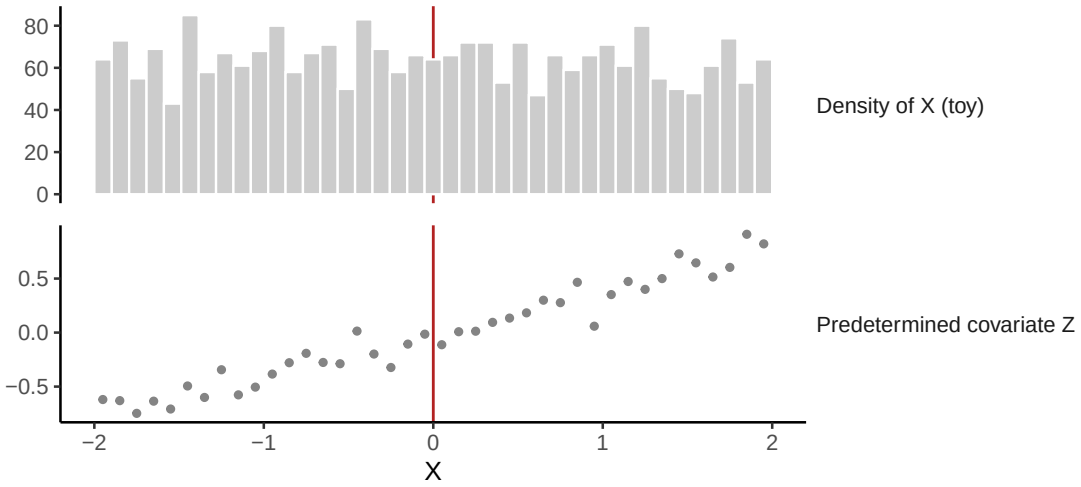
Diagnostics: what to check (Imbens–Lemieux)

- **Manipulation/sorting:** does the density of X jump at c? (McCrary test)
- **Covariate continuity:** do predetermined covariates jump at c?
- **Placebo cutoffs:** do you see similar jumps at other values?

Interpretation: these checks probe whether "just below vs. just above c" is a credible comparison.

Toy RD diagnostics: density and covariate balance

In the toy DGP there is no manipulation by construction, and the extra covariate Z is smooth in X.



RD in practice: general recommendations (Imbens–Lemieux style)

Local linear within a bandwidth: estimate the jump at the cutoff by OLS on units with $|X_i - c| < h$:

$$Y_i = \alpha_0 + \tau \cdot D_i + \beta_1(X_i - c) + \beta_2 D_i(X_i - c) + \varepsilon_i.$$

With a uniform (rectangular) kernel, this is local linear regression at a boundary; $\hat{\tau}$ estimates the cutoff jump.

- **Bandwidth choice:** use a data-driven "optimal" bandwidth (Imbens and Kalyanaraman, 2012), but still report sensitivity across multiple h's.
- **Inference:** use robust standard errors (or the bootstrap); show the estimated CEF within the bandwidth.
- **Robustness:** report estimates varying polynomial order/specification and bandwidth.
- **Diagnostics:** test continuity of predetermined covariates and of the density/PDF of X at c (McCrary-type check).

RD robustness: specification tests (what to report)

Assess robustness by checking whether the "local comparison" story is supported in the data:

- **Covariate continuity:** look for jumps in predetermined covariates at c.
- **Density continuity:** test for discontinuities in the conditional density of X at c (sorting/manipulation).
- **Placebo cutoffs:** check whether $\mathrm{E}[Y \mid X = x]$ shows discontinuities at other values of x.
- **Bandwidth + specification sensitivity:** vary h and the functional form, and report estimates *with and without* additional covariates (as precision adjustment).

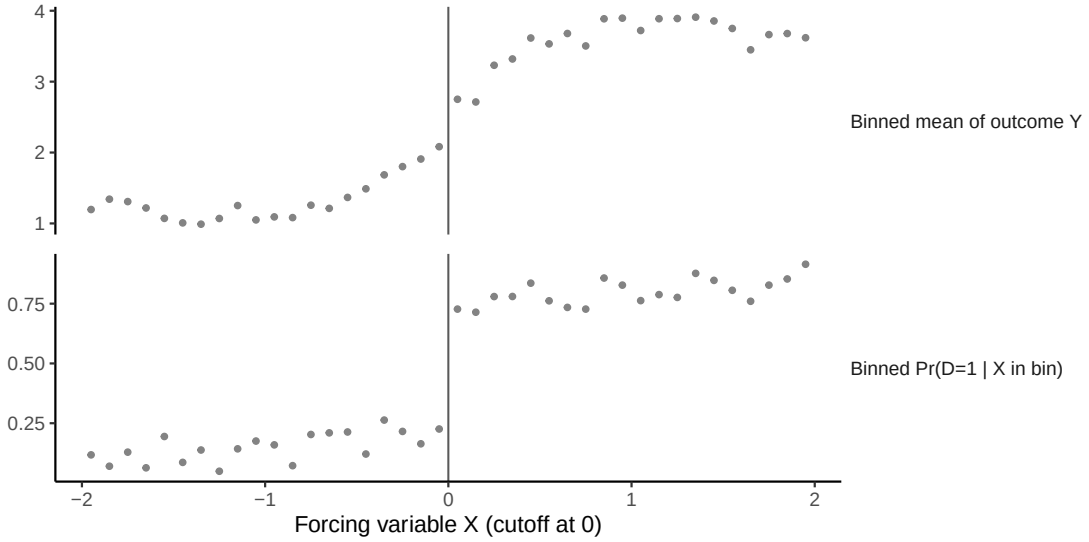
Case 2: fuzzy RD (FRD) designs — practical workflow

FRD adds one key complication: treatment receipt D_i is not perfectly determined by eligibility $Z_i = \mathbf{1}\{X_i \geq c\}$.

- **Graphs:** make the binned outcome plot *and* graph $\Pr(D_i = 1 \mid X_i = x)$ (first stage) near c.
- **Estimation:** estimate τ_{FRD} using TSLS (numerically equivalent to the local Wald ratio Δ_Y / Δ_D).
- **Inference:** use robust TSLS standard errors (or bootstrap); a plug-in approach is another option.
- **Bandwidth:** choose an "optimal" bandwidth as a starting point (IK 2012 / modified CV ideas), and still report sensitivity.
- **Robustness:** apply the SRD specification tests; also compare the FRD estimate to standard unconfoundedness-based estimates as a sanity check.

Toy FRD: binned outcomes and treatment probability

In FRD, it is helpful to plot the reduced-form jump in Y *and* the first-stage jump in D.



Fuzzy RD: the local Wald estimand

Define outcome and treatment jumps at the cutoff:

$$\Delta_Y = \lim_{x \downarrow c} \mathrm{E}\left[Y \mid X = x\right] - \lim_{x \uparrow c} \mathrm{E}\left[Y \mid X = x\right],$$

$$\Delta_D = \lim_{x \downarrow c} \mathrm{E}\left[D \mid X = x\right] - \lim_{x \uparrow c} \mathrm{E}\left[D \mid X = x\right].$$

Then the fuzzy RD estimand is the local Wald ratio:

$$\tau_{FRD} = \frac{\Delta_Y}{\Delta_D}.$$

Equivalently, written directly in terms of the limits:

$$\tau_{FRD} = \frac{\lim_{x \downarrow c} \mathrm{E}\left[Y \mid X = x\right] - \lim_{x \uparrow c} \mathrm{E}\left[Y \mid X = x\right]}{\lim_{x \downarrow c} \mathrm{E}\left[D \mid X = x\right] - \lim_{x \uparrow c} \mathrm{E}\left[D \mid X = x\right]}.$$

Note: in FRD, treatment receipt can be endogenous, so it is *not* generally true that

$$\mathrm{E}\left[Y_i(d) \mid D_i = d,\ X_i = x\right] = \mathrm{E}\left[Y_i(d) \mid X_i = x\right].$$

This equality would follow under an unconfoundedness-type assumption, but FRD identification of causal effects does not require it.

Under additional assumptions (monotonicity + exclusion-type conditions), τ_{FRD} is a LATE for compliers at the cutoff:

$$\tau_{FRD} = \mathrm{E}\left[Y_i(1) - Y_i(0) \mid \text{unit } i \text{ is a complier},\ X_i = c\right].$$

FRD monotonicity and compliance types (RD analog of IV)

To interpret FRD as a LATE, it is useful to define potential treatment status under *counterfactual cutoff values*.

- Let $D_i(c') \in \{0, 1\}$ denote whether unit i would be treated if the cutoff were set to c' (keeping the unit's forcing value X_i fixed).

- **Monotonicity at the cutoff (Imbens–Lemieux):** $D_i(c')$ is non-increasing in c' at $c' = c$.

Define **compliance status** by how $D_i(c')$ changes as the cutoff crosses the unit's X_i:

- **Complier:** $\lim_{c' \downarrow X_i} D_i(c') = 0$ and $\lim_{c' \uparrow X_i} D_i(c') = 1$.
- **Never-taker:** $\lim_{c' \downarrow X_i} D_i(c') = 0$ and $\lim_{c' \uparrow X_i} D_i(c') = 0$.
- **Always-taker:** $\lim_{c' \downarrow X_i} D_i(c') = 1$ and $\lim_{c' \uparrow X_i} D_i(c') = 1$.
- **Defier:** $\lim_{c' \downarrow X_i} D_i(c') = 1$ and $\lim_{c' \uparrow X_i} D_i(c') = 0$.

Interpretation (with the usual assignment direction $D = 1\{X \geq c\}$): if the cutoff is set just *below* X_i, unit i is treated; if the cutoff is set just *above* X_i, unit i is not treated. Under monotonicity, there are no "defiers" (units whose treatment would increase when the cutoff is raised).

Next: differences in differences (DiD) and panel data

DiD setup: two groups, two periods

DiD uses changes over time plus a comparison group.

- Groups: $G_i \in \{0, 1\}$ (1 = treated group, 0 = comparison group)
- Time: $t \in \{0, 1\}$ (0 = pre, 1 = post)
- Treatment: $D_{it} = \mathbf{1}\{G_i = 1\} \cdot \mathbf{1}\{t = 1\}$ (2x2 case)
- Potential outcomes: $Y_{it}(1)$, $Y_{it}(0)$, observed $Y_{it} = Y_{it}(D_{it})$

2x2 DiD estimand

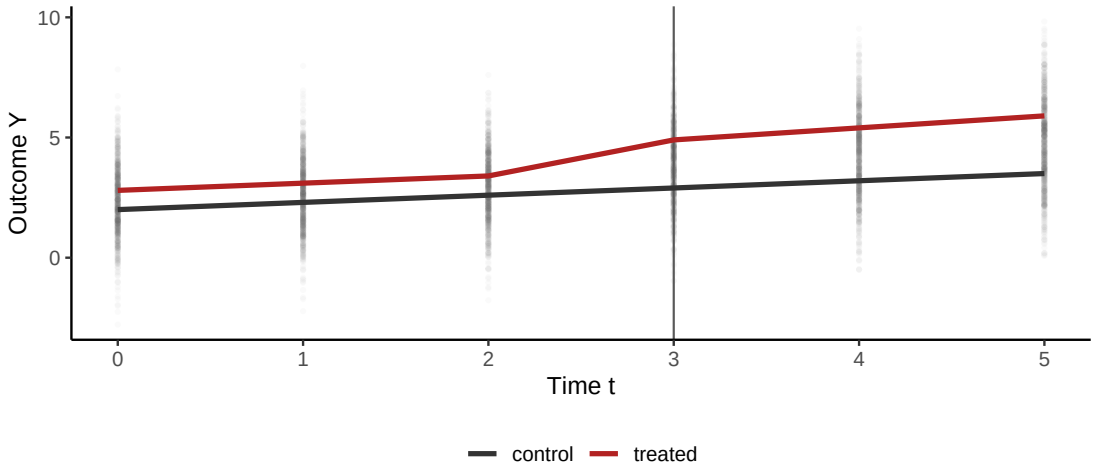
The DiD estimand is the difference in average changes:

$$\tau_{DiD} \equiv \Big(\mathrm{E}\left[Y \mid G = 1, t = 1\right] - \mathrm{E}\left[Y \mid G = 1, t = 0\right]\Big) - \Big(\mathrm{E}\left[Y \mid G = 0, t = 1\right] - \mathrm{E}\left[Y \mid G = 0, t = 0\right]\Big).$$

Interpretation: remove (i) time shocks common to both groups and (ii) time-invariant level differences between groups.

DiD picture: noisy raw data vs. group means

Raw panel data are noisy; DiD is about *average* changes (thick lines show the expected group means in a toy DGP with parallel pre-trends).



2x2 DiD as a regression

In the 2x2 case, the DiD estimator equals the coefficient on the interaction term in:

$$Y_{it} = \alpha + \gamma\, G_i + \lambda\, \text{Post}_t + \tau\,(G_i \cdot \text{Post}_t) + \varepsilon_{it},$$

where $\text{Post}_t \equiv \mathbf{1}\{t = 1\}$.

- γ: baseline level difference (treated vs comparison)
- λ: common time shock (post vs pre)
- τ: DiD estimate of the treatment effect under parallel trends

Adding covariates is typically for precision (and for making parallel trends more plausible), but it does not replace the design assumption.

Parallel trends (the identification assumption)

The key design assumption is about *counterfactual* trends under no treatment:

$$\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1\right] = \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0\right].$$

One formal (strong) "common trends" condition: letting $\Delta Y_i(d) \equiv Y_{i1}(d) - Y_{i0}(d)$, assume

$$\big(\Delta Y_i(0), \Delta Y_i(1)\big) \perp (D_{i0}, D_{i1}).$$

This implies the mean parallel-trends condition above, but it is stronger than what is needed for identification.

Positivity/overlap (2x2): $\Pr(G_i = 1) \in (0, 1)$, and we observe both groups in both periods so each cell mean $\mathrm{E}\left[Y \mid G = g, t = t\right]$ is well-defined.

Under parallel trends (and SUTVA), τ_{DiD} identifies the average treatment effect for the treated group in the post period:

$$\tau_{DiD} = \mathrm{E}\left[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1\right].$$

What can go wrong? (threats to parallel trends)

Parallel trends is a claim about the *untreated counterfactual* for the treated group.

- **Differential shocks:** treated group faces a shock in the post period unrelated to treatment.
- **Composition changes:** who is observed in each group changes over time.
- **Anticipation:** treatment affects behavior before the official start.
- **Spillovers:** the comparison group is affected indirectly.

Pre-trends: event-study leads are useful diagnostics, but a flat pre-trend is not a proof (power is often limited).

Inference: serial correlation (Bertrand–Duflo–Mullainathan 2004)

In DiD with repeated observations over time, errors are often serially correlated.

- Naive i.i.d. standard errors can be much too small.
- Practical default: cluster standard errors at the unit level (or at the level of treatment assignment).
- With few clusters: be cautious (cluster-robust SEs can be unreliable).

Key message: inference is part of the design – not an afterthought.

What to cluster on? (rule of thumb)

- Cluster at the level of **treatment assignment / policy variation**.
- If treatment varies at the unit level over time, unit-level clustering is a common default.
- If treatment varies at a higher level (e.g., state policy), cluster at that higher level (often two-way with time if needed).
- With **few clusters**: consider alternatives (aggregation, randomization-style inference, or methods designed for small G).

From 2x2 to many periods: TWFE

With panel data, a common starting point is the two-way fixed effects (TWFE) regression:

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}.$$

- If there is one treated group and one treatment time, TWFE aligns with 2x2 DiD.
- With staggered adoption and heterogeneous effects, TWFE can mix many comparisons in hard-to-interpret ways.

Event-study regression (multi-period)

With staggered adoption, a common goal is a dynamic effect profile by event time $k = t - G_i$.
One regression-style approach is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \, \mathbf{1}\{t - G_i = k\} + \varepsilon_{it},$$

where $k = -1$ (one pre period) is omitted as the baseline.

Interpretation: β_k are "effects at event time k" *if* the regression targets a clean estimand (this can fail under TWFE with heterogeneous effects).

Staggered adoption: what can go wrong with TWFE?

With multiple cohorts and adoption times, TWFE implicitly averages many 2x2 comparisons.

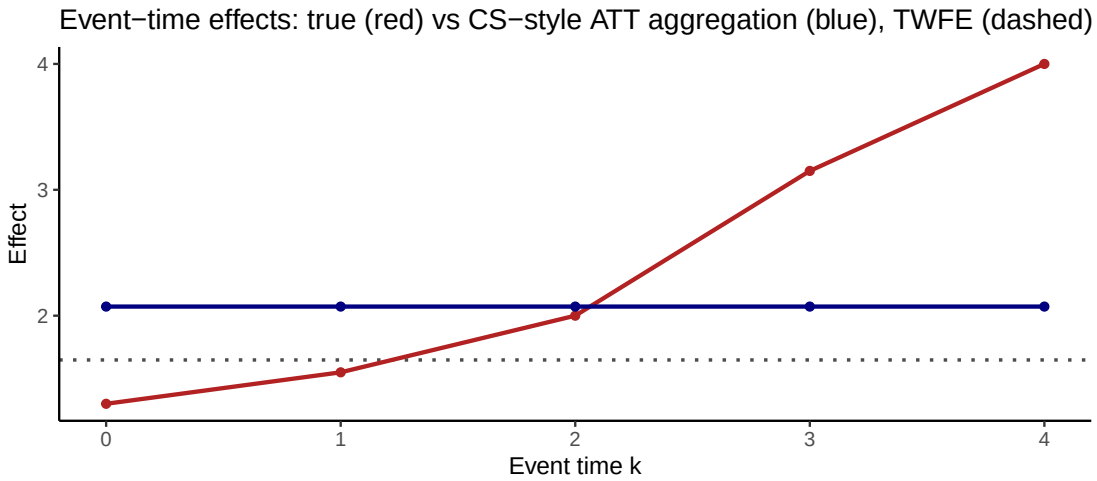
- Some comparisons use already-treated units as controls for later-treated units.

- Weights can be non-intuitive (and can be negative), so coefficients may not equal the average of cohort-specific effects.

See Goodman-Bacon (2021) for the decomposition intuition; see Sun & Abraham (2021) for event-study issues and fixes.

Toy staggered adoption: TWFE vs. transparent targets

In a simple staggered-adoption toy model with heterogeneous effects, TWFE can differ from a transparent "average of cohort-time effects."



Toy message: when effects differ by cohort/event time, define the estimand first (e.g., $ATT(g, t)$ or event-time averages), then use an estimator that targets it.

Modern targets: group-time effects

One clean estimand with staggered adoption is the group-time ATT:

$$ATT(g,t) \equiv \mathrm{E}\left[Y_{it}(1) - Y_{it}(0) \mid G_i = g\right], \qquad t \geq g,$$

where G_i is the first-treated time for unit i.

Modern estimators (e.g., Callaway–Sant'Anna) build event-study plots by aggregating $ATT(g,t)$ across cohorts with transparent weights.

Callaway–Sant'Anna: comparison sets and aggregation

For each treated cohort g and time $t \geq g$, define a **control set** $C_{g,t}$:

- **Never-treated controls:** units that are never treated.
- **Not-yet-treated controls:** units with $G_i > t$ (untreated by time t).

One (simplified) way to think about $ATT(g,t)$ is a 2x2 DiD within cohort g, using baseline $g-1$:

$$ATT(g,t) \approx \Big(\mathrm{E}\left[Y_t - Y_{g-1} \mid G = g\right]\Big) - \Big(\mathrm{E}\left[Y_t - Y_{g-1} \mid i \in C_{g,t}\right]\Big).$$

Aggregation: to plot event time $k = t - g$, average $ATT(g, g+k)$ across cohorts g using transparent weights (often proportional to cohort size or share treated at that event time).

Modern workflow (high level)

With staggered adoption, a useful workflow is:

1. Define the estimand: $ATT(g, t)$ (or event-time effects) for clear g, t.
2. Choose a comparison group: never-treated or not-yet-treated units.
3. Estimate $ATT(g, t)$ within each cohort-time cell.
4. Aggregate to event time $k = t - g$ with transparent weights; plot leads/lags.
5. Do robustness checks (windows, alternative controls, placebo outcomes).

Callaway–Sant'Anna (2021) and Sun–Abraham (2021) are standard references for modern practice.

Event studies: what we want to see

An event-study plot shows average effects by event time $k = t - g$.

- Pre-period coefficients (leads) are a diagnostic for trend violations/anticipation.

- Post-period coefficients (lags) summarize dynamic treatment effects.

Caution: a good-looking pre-trend plot is helpful, but it is not a proof of parallel trends.

Robustness checklist (DiD)

- Vary the event window and sample restrictions (who is included as treated/control?).
- Try alternative comparison groups (never-treated vs not-yet-treated).
- Check sensitivity to covariate adjustment (precision vs identifying variation).
- Placebos: outcomes that should not respond; or placebo policy timing.
- Inference sensitivity: clustering level; few-cluster concerns; alternative SE methods.

Synthetic control (SC): when DiD is too thin

When there is a single treated unit (or a small number) and a long time series, DiD with one comparison unit can be fragile.

- Synthetic control constructs a weighted combination of control units to match the treated unit's pre-treatment outcomes.

- Then it uses that weighted combination as the counterfactual post-treatment trend.

Core references: Abadie & Gardeazabal (2003), Abadie, Diamond & Hainmueller (2010).

SC vs. DiD (conceptual link)

- Both are about constructing a credible counterfactual trend.
- DiD: a comparison group + a parallel-trends assumption.
- SC: a data-driven comparison group (weights) chosen to match the pre-period.

Tradeoff: SC can improve pre-period fit, but it does not eliminate the need for a design argument (and it is still local/design-dependent).

Week 8 summary (RD + DiD)

- RD: under continuity at the cutoff, the jump in $\mathrm{E}[Y \mid X = x]$ identifies $\mathrm{E}[Y(1) - Y(0) \mid X = c]$ (local effect).
- DiD: under parallel trends, the difference in average changes identifies a causal effect for the treated group.
- With panels: inference is hard (serial correlation); cluster appropriately.
- With staggered adoption: define the estimand first; don't default to TWFE.

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